Problem 04: Real-Time Multilingual Conversational AI Tutor with Adaptive Learning and Knowledge Tracing
Domain: EdTech | NLP | Speech AI | Reinforcement Learning
Difficulty: Medium-Hard to Hard

Problem
Static learning material and uniform tests ignore each student's gaps, language, pace and recurring misconceptions.
Objective
Build a conversational tutor that teaches in multiple languages, assesses understanding, finds knowledge gaps and adapts later lessons.
Users
Students (school/college), teachers.
Core Requirements

- Text plus at least one more modality (voice).
- English plus at least one Indian language.
- Searchable knowledge base from curated material.
- Grounded explanations with source references.
- Quiz generation and answer evaluation.
- Per-concept mastery tracking.
- Recurring-misconception detection.
- Difficulty adjustment from performance.
- Revision topics and personalized learning paths.
- Teacher dashboard with concept-level progress.

Advanced Challenges

- Knowledge tracing: BKT, DKT or mastery models.
- Retrieval-grounded question generation.
- Multilingual ASR and TTS.
- Separating understanding from memorization.
- Adaptive next-question/lesson selection (bandit or RL).
- Measuring learning gain, not only satisfaction.

Suggested Datasets
NCERT textbooks (open), Wikipedia/Khan-style content, ASSISTments and EdNet (for knowledge tracing), AI4Bharat IndicTrans2 / IndicWhisper / IndicTTS resources, FLORES for language eval.
Suggested Tech
Python, Whisper, multilingual embeddings (e.g., multilingual-e5, LaBSE), RAG, Hugging Face, pyBKT, Streamlit/React, FastAPI.
Reference Architecture
Voice/text input → ASR → language ID → retriever (curriculum KB) → tutor LLM with citations → answer evaluator → knowledge-tracing engine → next-item policy → TTS; teacher analytics store.
Evaluation
Answer correctness, citation faithfulness, language-specific accuracy, knowledge-tracing AUC on held-out students, pre/post assessment gain, ASR WER per language.
Expected Deliverable
Adaptive tutor with multilingual chat/voice, dynamic quizzes, mastery estimates, misconception flags, revision plans, teacher analytics.

Pitfalls
Poor quality in the Indian language TTS/ASR; evaluating only on simulated students without saying so.
Suggested Timeline (48h)
0-10h KB + RAG; 10-22h multilingual + voice; 22-34h quiz + knowledge tracing; 34-42h adaptivity; 42-48h teacher dashboard.

Standard Judging Rubric (applies to every problem)

Problem understanding	10%
Technical depth	20%
Data engineering (quality, leakage prevention, reproducibility)	10%
Model performance (metrics, baselines, honest evaluation)	15%
System architecture	15%
Working prototype	10%
Explainability and responsible AI	10%
Innovation and scalability	5%
Presentation and demo	5%

Mandatory Submission Checklist

- □ GitHub repository with setup instructions
- □ Working prototype or recorded demo
- □ Documented system architecture
- □ Dataset sources and preprocessing methodology
- □ Evaluation results with baseline comparisons
- □ Limitations and failure cases
- □ One technically challenging feature demonstrated
- □ Short presentation: problem, approach, results, future work